

Chemistry

12

MCQs:-

* 6
22

1. C

2. C

3. D

4. B

5. B

6. A

7. A

8. C

9. B

10. B

11. A

12. D

13. D

14. A

15. D

16. A

17. C

18. B

19. D

20. B

21. B

22. ~~C~~ C

23. A

24. D

25. D

26. C

27. ~~D~~ D

28. C

29. B

30. A

31. B

32. A

33. B

34. ~~A~~ B

35. B

36. C

37. A

38. C

39. D

40. C

41. C

42. D

43. A

44. B

45. D

46. B

47. C

48. D

49. A

50. B

51. A

52. A

53. A

54. B

55. B

56. D

57. ~~B~~ B

58. B

59. B

60. C

Q.2(i) —
assuming mass 100g
A. 2(i) —

$$N = \frac{30.4}{14} = 2.17$$

$$O = \frac{69.6}{16} = 4.35$$

now division by smallest value.

$$N = \frac{2.17}{2.17} = 1$$

$$O = \frac{4.35}{2.17} \approx 2$$

Ratio = 1:2

Empirical formula = NO₂

(b) molecular formula:-

$$14 + 2(x \cdot 16) = 92$$

molecular mass = 92

$$\frac{92}{46} = 2$$

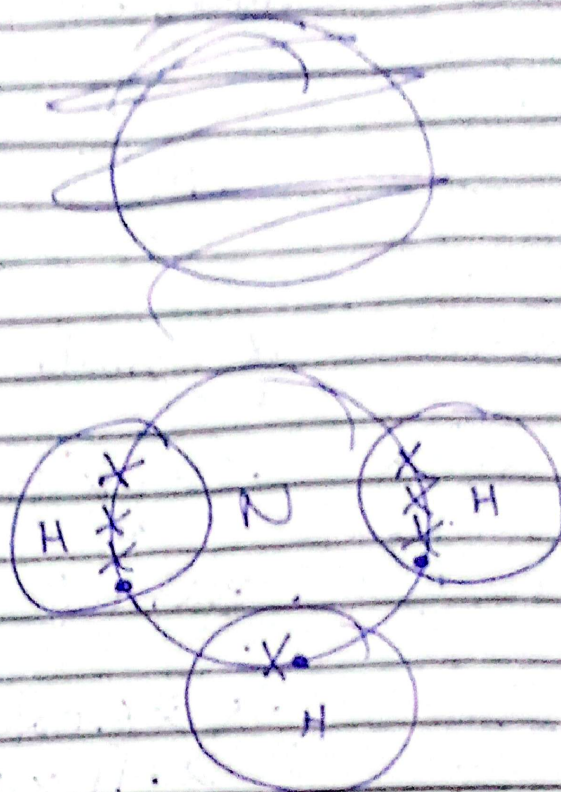
$$46$$

multiplying by 2:-
N₂O₄

Q: 2(ii)

A: 2(ii)

(a)



Bond pairs : 3

Lone pairs : 1

Q: 3(i)

A: 3(i)

(a) formula mass of K_2SO_4

$$2(39) + 32 + 4(16)$$

$$78 + 32 + 64 = 174 \text{ amu.}$$

② moles in 60.9g

$$\text{moles} = \frac{\text{Mass}}{\text{formula mass}}$$

$$\frac{60.9}{174} = 0.35 \text{ mol}$$

③ number of formula units:-

$$\text{moles} = 6.022 \times 10^{23}$$

$$\text{NA} = 6.022 \times 10^{23}$$

$$\text{moles} = 0.35 \text{ mol}$$

$$\text{f. u.} = \frac{0.35 \times 6.022 \times 10^{23}}{2.11 \times 10^{23}}$$

